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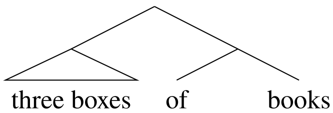
Recursion in NP: pseudopartitive measures require complementation, not specification

Malhaar Shah

1. Introduction

I aim here to provide evidence favoring one of two hypotheses regarding the structure of measure phrases in pseudopartitives (*three boxes of books*). I will call hypothesis-(1a) the ‘Spec’ hypothesis (Schwarzschild 2006), and hypothesis-(1b) the ‘Comp’ hypothesis (Stickney 2009).

(1) a.



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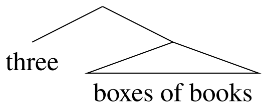

This paper has two claims:

i) The semantic motivations provided for the Spec hypothesis do not force an implementation using the structure in (1a).

ii) The Comp hypothesis, and not the Spec hypothesis, is consistent with the results of syntactic diagnostics (including facts about agreement, constituency, phrasality, and locality).

I begin with reviewing the motivation for the Spec hypothesis. I argue that it does not show what is required to decide between (1a) and (1b). I then outline the predictions of each hypothesis across four relevant syntactic dimensions, and show that only the predictions of the Comp hypothesis are borne out. This means that the semantic phenomena used to motivate the Spec hypothesis are ought to be reanalyzed without appeal to the syntactic implementation in (1a).

I end by drawing out a theoretical consequence: if the Comp hypothesis it correct, it requires nominal complements to nominals (recursion inside NP). This has been more controversial than clausal recursion (Stowell 1989; Frank & Kroch 1994; Frank 2002; Chomsky 2008; Kayne 2008; Citko 2014), but if the Comp hypothesis is correct, then recursion inside NP must be allowed.

2. The motivation for the Spec hypothesis

Schwarzschild (2006) provides a semantic argument for the Spec hypothesis. It allows for a straightforward implementation of a semantic generalization, known as Monotonicity.

2.1. Monotonicity

The intuition behind monotonicity is that there is a semantic relation between the thing being measured and the measure phrase. The choice of dimension (volume, weight, temperature) must satisfy a constraint

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which has named ‘monotonicity’. That is, we can use natural language measures only if the relationship between them and the substance measured (denoted by the ‘substance’ noun) is ‘monotonic’:

(2) **Monotonicity, informally:** a measure function μ (a mapping from Dom_μ to a unique degree) is monotonic iff, for any $x, y \in Dom_\mu$, y is a proper part of $x \leftrightarrow \mu(y) < \mu(x)$

A measure is monotonic iff its value for x is always higher than its value for a proper part of x.

This places a condition on measure phrases:

(3) a. two litres of soup (volume is monotonic)

b. #two degrees of soup (temperature is non-monotonic)

c. two degrees of global warming

The reasoning ruling (3b) out goes as follows. Take a bowl of soup, and call it Campbell. Any larger bowl of soup into which you pour Campbell will also increase along the volume dimension. By having Campbell as a proper part, the larger bowl of soup must also have a greater volume. Volume is then is monotonic on the substance, soup.

But if Campbell is 90°C (194°F), and Campbell is poured into a larger bowl of soup, there is no saying whether the larger bowl of soup is of a greater or lesser temperature, despite containing Campbell as a proper part. Temperature, unlike volume, is non-monotonic on the substance. As only dimensions that are monotonic measures on the measured substance are allowed to act as measure phrases, we explain the unavailability of (3b).

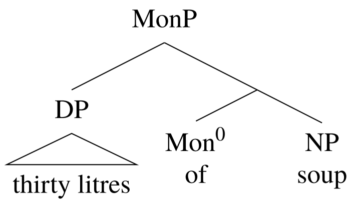
Importantly, this constraint does *not* turn up in ordinary NP-modification, meaning that the syntax/semantics of the pseudopartitive is involved:

(4) a. four inches of cable (length only)

b. four-inch cable (width; non-monotonic on cable)

Schwarzschild encodes this as stemming from the lexical semantics of a functional head which is present in pseudopartitives, which he calls Mon^0. It takes as its complement the noun denoting the substance, and as its specifier, the phrase denoting the measure.

(5)



He draws an analogy to Voice^0 in the verbal domain (Kratzer 1996), which assigns a agent θ-role to its Specifier. Similarly, Mon^0 assigns a measure θ-role to its Specifier, imposing monotonicity.

2.2. Monotonicity in verbal comparatives

Schwarzschild’s theory captures the Monotonicity by treating it as a relation imposed on ⟦Spec, MonP⟧ and ⟦Comp, MonP⟧ by the lexical semantics of the functional head Mon^0. This functional head, Schwarzschild hypothesizes to be a member of the nominal Extended Projection.

However, the monotonicity constraint appears in not only in pseudopartitives, but in comparatives too. Importantly, it appears in *verbal* comparatives, where it seems harder to locate Mon^0 (which is a member of only the nominal Extended Projection, in Schwarzschild's theory).

(6) Mal drank [$_{\mathrm{NP}}$ more coffee than Val]. (✓volume, ✗temperature, ✗intensity)

(7) Val [$_{\mathrm{VP}}$ ran more than Al]. (✓duration, ✓distance, ✗intensity)

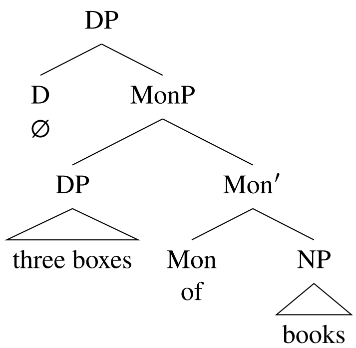
If the intensity of coffee is disallowed as the measure function of *more* because intensity is non-monotonic on coffee, it is tempting to say the intensity of a run is disallowed as the measure function of *more* because intensity is non-monotonic on runs. So it seems that *more* is monotonic whether it modifies *coffee*$_{\mathrm{NP}}$ or *run*$_{\mathrm{VP}}$. But then it is implausible that a functional head in the NP Extended Projection, pronouncable as *of*, is the source of monotonicity.

This is at least a *prima facie* reason to doubt Schwarzschild's MonP-based explanation of his significant observations, undercutting the argument from Monotonicity for the Spec-hypothesis.

3. Syntactic tests favor the Comp hypothesis

Having seen reason to doubt the motivation for the Spec hypothesis, we now turn to independent predictions of the hypotheses. I assume Schwarzschild's implementation of the Spec hypothesis, where the category Mon serves as the label of the whole phrase:

(8)



This category has gone by different names (e.g., MeasP; Solt 2015; Wilson 2018), but as long as the measure noun [$_{\mathrm{DP}}$ *three boxes*] sits in the specifier of the phrase headed by *of*, the predictions are as I argue below.

1. **Constituency:** The Spec hypothesis predicts that *three boxes* should behave as a constituent.
2. **Agreement:** The Spec hypothesis predicts agreement with the measured noun.
3. **Phrasality:** The Spec hypothesis predicts [*of books*] is not an XP, and so, cannot move or delete.
4. **Locality:** The Spec hypothesis predicts adding measure phrases does not deepen measured nouns.

3.1. Predictions of the hypotheses in more detail

Firstly, the Spec hypothesis requires [$_{\mathrm{DP}}$ 99 bottles] to be a constituent, and should be able to move or delete like a constituent (modulo left-branch effects, Ross 1967). The Comp hypothesis predicts that [$_{\mathrm{DP}}$ 99 bottles] can behave as a constituent, but only in a remnant movement configuration, where [$_{\mathrm{DP}}$ of beer] has moved, as in [$_{\mathrm{DP}}$ 99 [bottles t_{PP}]].

Secondly, the Spec hypothesis and the Comp hypothesis differ in their predictions about which noun controls subject-verb agreement. If the probing T Agrees with the closest c-commanded NP (Chomsky 2000), then the Spec-hypothesis predicts that the substance noun controls agreement. The Comp-hypothesis, on the other hand, predicts that the measured noun controls agreement.

(9) a. [$_{\text{TP}}$ T [$_{v\text{P}}$ [$_{\text{DP}}$ [$_{\text{MonP}}$ [$_{\text{DP}}$ 99 bottles] of **beer**]] [v VP]]] (Spec: predicts SG agreement)

b. [$_{\text{TP}}$ T [$_{v\text{P}}$ [$_{\text{DP}}$ 99 [$_{\text{NP}}$ **bottles** [$_{\text{PP}}$ of beer]]] [v VP]]] (Comp: predicts PL agreement)

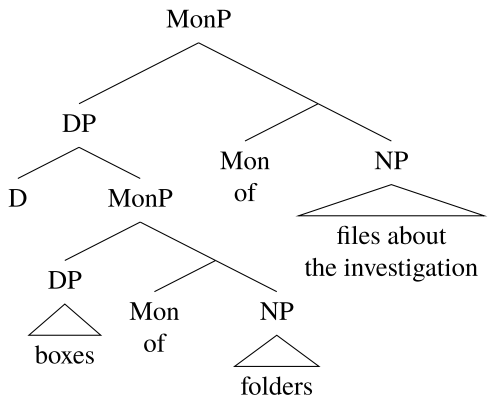
Thirdly, *of beer* should not behave as a phrasal constituent, at least according to the implementation of the Spec hypothesis in Schwarzschild (2006). The node containing those lexical items is a bar level Mon′, and therefore, should not be able to move or delete (under the conventional assumption that movement and ellipsis target only XPs, and not X′s).

Fourthly and finally, the hypotheses make different predictions about locality of movement. Both hypotheses allow for iteration of measure phrases inside the pseudopartitive:

(10) The shelves of (boxes of) (folders of) (...) files about the investigation

The Spec hypothesis seems to require a highly complex specifier for these cases:

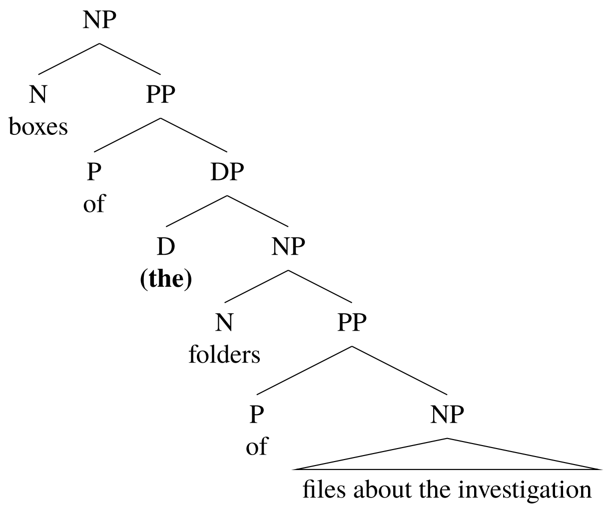
(11)



This means that any of the iterated measure phrases do not contribute to the depth of embedding of the NP [$_{\text{NP}}$ files about the investigation]. Any properties of the iterated measure phrases (like definiteness) cannot contribute to the grammaticality of extraction from [$_{\text{NP}}$ files about the investigation].

On the other hand, under the Comp hypothesis, if *folders* were definite, extraction from [$_{\text{NP}}$ files about the investigation] *would* count as extraction out of a definite:

(12)



Given previous observations that extraction out of definites is worse than indefinites (see, e.g., Davies & Dubinsky 2003), then the Comp hypothesis predicts that the definiteness of iterated meaasure phrases should degrade extraction from the rightmost substance noun, but the Spec hypothesis predicts no difference in grammaticality.

The observed data are entirely consistent with the syntax in the Comp hypothesis, but not the syntax in the Spec hypothesis.

3.2. Results of the tests

Firstly, the tests for constituency of the measure phrase are somewhat equivocal. Wilson (2018) reports the following judgments, at least with focus as indicated:

(13) It was [a small handful] that he ate of walnuts.

(14) Three generous teaspoons, she added, of MOLASSES.

However, judgments here (including – but not limited to – my own) vary. Furthermore, given that English generally allows the extraposition of prepositional phrases, these facts do provide evidence for the Spec hypothesis over the Comp hypothesis, as the relevant sentences can be generated under the Comp hypothesis by remnant movement:

(15) [Three generous [teaspoons t_1]]$_2$ [she added t_2] [$_{\text{PP}}$ of molasses]$_1$

So, on their own, these constituency tests do not provide evidence for the Specifier-status of a measure phrase (*contra* Wilson 2018).

Secondly, we see Agreement with a plural measure phrase, not the singular substance noun.

(16) There {are/*is} 99 **bottles** of beer on the wall.

This is consistent only with the Comp hypothesis (assuming Agree finds the closest c-commanded noun, not merely leftward search – cf. Kayne 2011, p. 12).

Thirdly, we find that substance nouns do behave like maximal projections, in that they can move and delete. Partitive phrases can generally move, as well as be elided (Gagnon 2013).

(17) a. [Of beer]$_1$, they drank [99 bottles t_1] yesterday. (movement)

b. They drank [99 bottles t_1] yesterday [of beer]$_1$.

(18) They drank [99 bottles [of beer]], but we drank [a barrel [~~of beer~~]]. (ellipsis)

If movvement and ellipsis are restricted to maximal projections, then this tells against the Spec hypothesis, which could only get *of beer* to behave as a maximal projection by first positing movement of *a barrel*. However, this movement is otherwise unmotivated (unlike the PP-extraposition preceding the remnant movement for (15). Thus this is strong evidence for the Comp hypothesis.

Lastly, we find that adding definiteness to an intermediate measure phrase does degrade the sentence, consistent with the view that it would be movement out of the definite DP.

If the definite DP *is* in a Specifier position, there is no problem with subextraction from NP:

(19) What$_1$ did you enjoy [$_{\text{DP}}$ [$_{\text{Spec}}$ **the** editor (of **the** college newspaper)]'s review of t_1

Definites in SPEC cause no problem for subextraction. (19) contrasts with what we observe for partitives:

(20) The journalist exposed the scandal...
[which$_1$ the government buried three boxes of (??**the**) folders of reports about t_1]

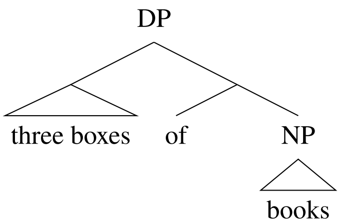
The definiteness of the *intervening* measure phrase can count for evaluating the acceptability of movement across it. This would follow if the movement in (20) was movement *out* of a definite (as required by the Comp-hypothesis; (12)), and not merely movement across a measure phrase (as in the Spec hypothesis; (11)). Thus we see additional syntactic evidence for the Comp hypothesis.

Across four syntactic tests, then, we see data that is consistent with the Spec hypothesis only in one test (where the judgments are somewhat hazy), and in that test, there are no results inconsistent with the Comp hypothesis. In the other tests, we see data consistent *only* with the Comp hypothesis. Thus, I submit that complementation is the way by which partitives are constructed, not specification. This has some interesting theoretical consequences.

4. Concluding remarks

This paper has argued that, regarding the syntax of pseudopartitive constructions, (1a) the 'Spec' hypothesis (Schwarzschild 2006) is incorrect, and (1b), the 'Comp' hypothesis (Stickney 2009), is correct.

(21) a.



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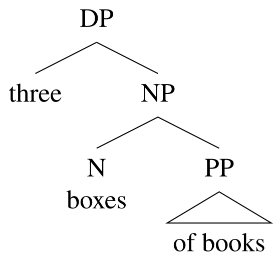

If this is right, then measure nouns must be able to take PP-arguments, which can themselves take DP arguments, which take NP arguments. This is much like verbs taking CP complements, which take TP complements, which take VP arguments. That is, there is recursion in NP.

This stands contrary to two theoretical claims, i) the claim that DPs do not allow recursion (Frank & Kroch 1994, Frank 2002, cf. Stowell 1989), and ii) the stronger claim that only verbs can take arguments (Kayne 2008). These theories must say that what appears to be a PP-complement to N is in fact a PP-argument of the verb (cf. Chomsky 1977 on PP-reanalysis). I cannot here test the implications of these analyses.

Furthermore, it would seem that another explanation of the Monotonicity generalization in pseudopartitives (Schwarzschild 2006) will be required. It cannot be a functional head in the nominal domain that is responsible for this.

Bresnan (1973) claimed that underlying "every comparative is a partitive or quantifier-like element". Both Wellwood and Schwarzschild build on this, with Wellwood (2019) positing an element MUCH in comparatives and Schwarzschild (2006, p. 89) noting that quantified partitives with *much*, *many*, *more*, and *most* also display Monotonicity. Exploring the connection between comparatives and partitives will surely yield a better understanding of the Monotonicity constraint, and hopefully in a way that is compatible with the syntactic facts observed in this paper.

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